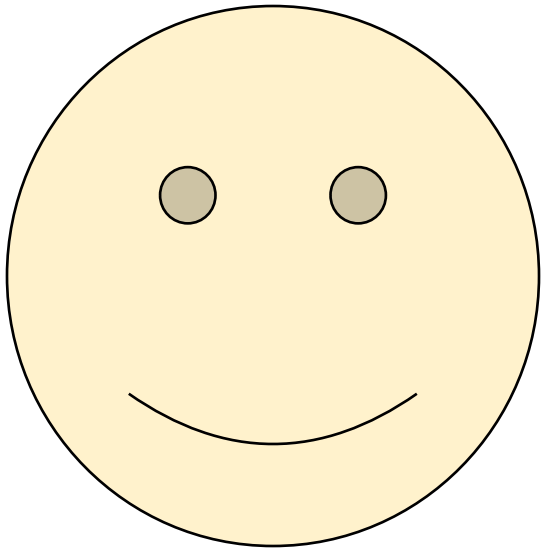
Pushing on Text Readability Assessment:
A Transformer Meets Handcrafted Linguistic Features

Bruce W. Lee[1,3], Yoo Sung Jang[2,3] and Jason Hyung-Jong Lee[3]

University of Pennsylvania[1], University of Wisconsin-Madison[2], LXPER AI[3]

Speaker: Bruce W. Lee

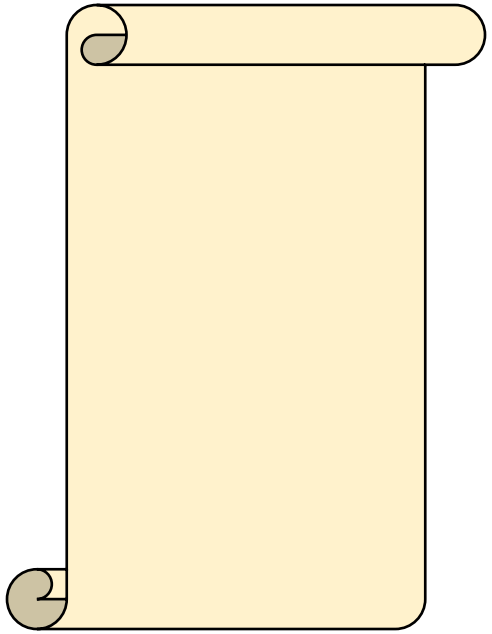
What is readability assessment?



Reader

Read

Some level of difficulty



Text/Article

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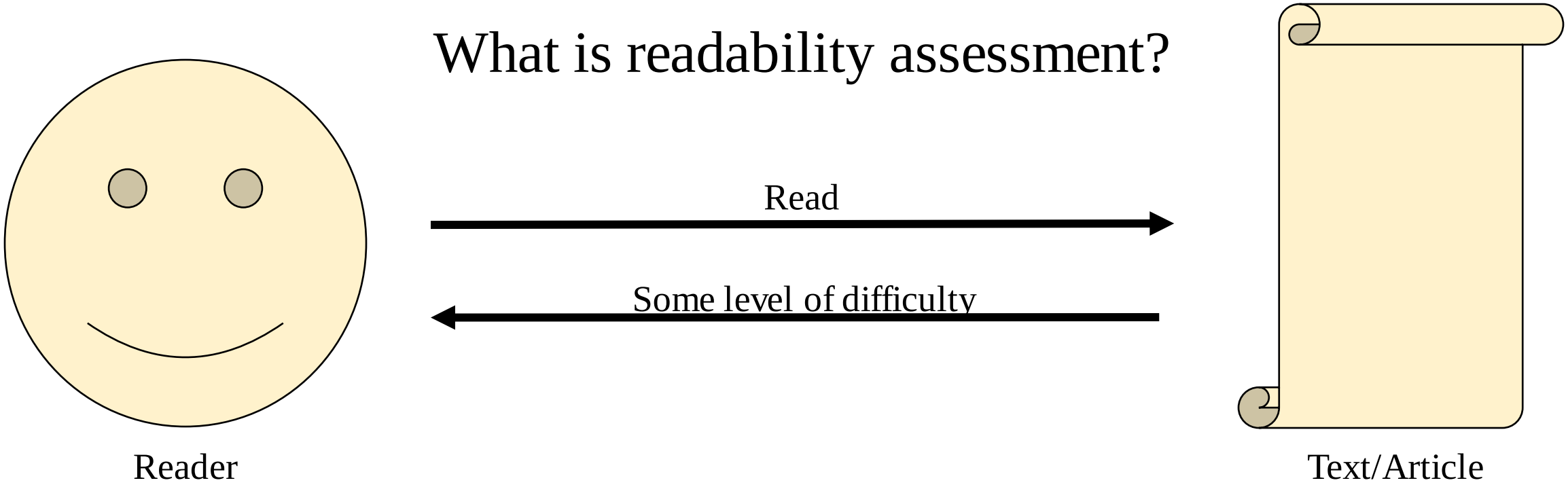
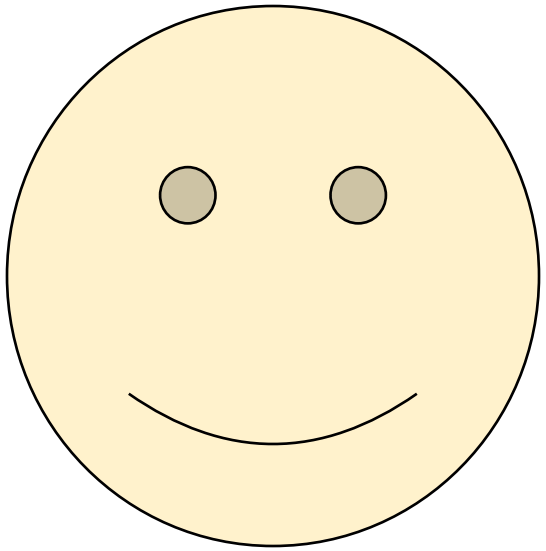
What is readability assessment?



1. Readability is the difficulty that a reader faces when reading a text.

2. One of the oldest systematic approaches to linguistics (Collins-Thompson, 2014).

What is readability assessment?



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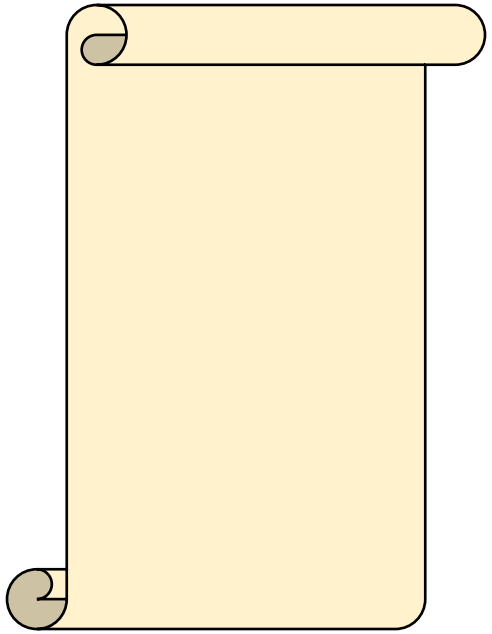

Text/Article

Linguistic Branches

Syntax

Semantic

Lexical

Discourse

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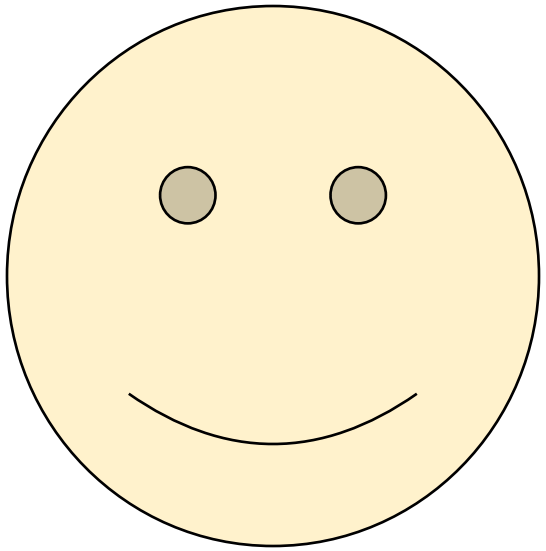

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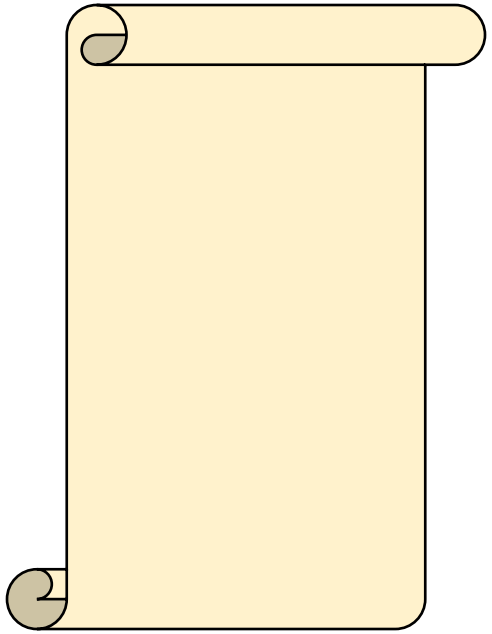

Text/Article

Linguistic Branches

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Handcrafted Linguistic Features

average count of noun phrases per sentence

ratio of subordinate clauses: verb phrases count

average parsed tree height per sentence

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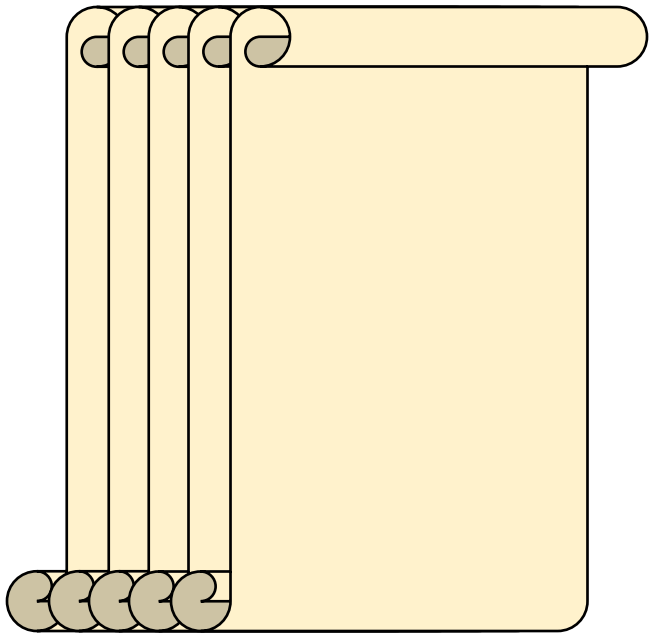

Readability-annotated dataset

Text	Readability Level
Abc motors has 3 cars and 1 truck but blah blah blah	3
...	5
...	1
...	2

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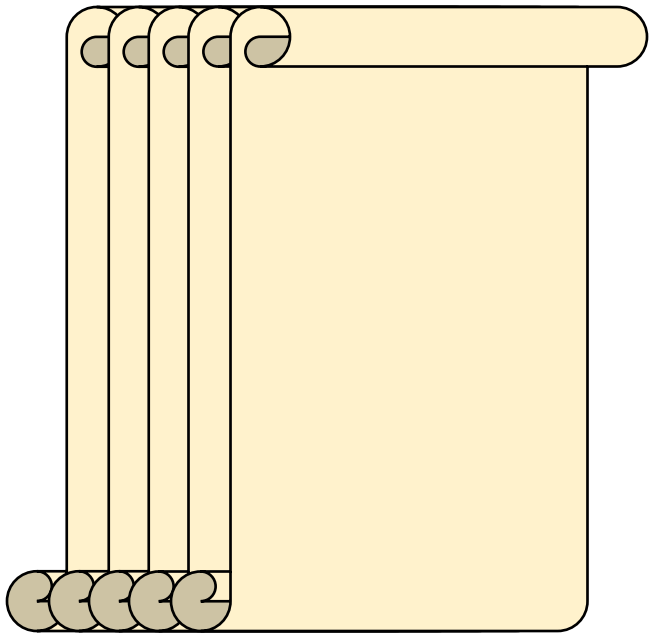
What is readability assessment?



Readability-annotated dataset

Text	Readability Level	Feature 1	Feature 2
Abc motors has 3 cars and 1 truck but blah blah blah	3	2.5	127
...	5	4.2	342
...	1	0.5	43
...	2	1.3	150

1. Feature study focused.

2. So many noteworthy linguistic features found.

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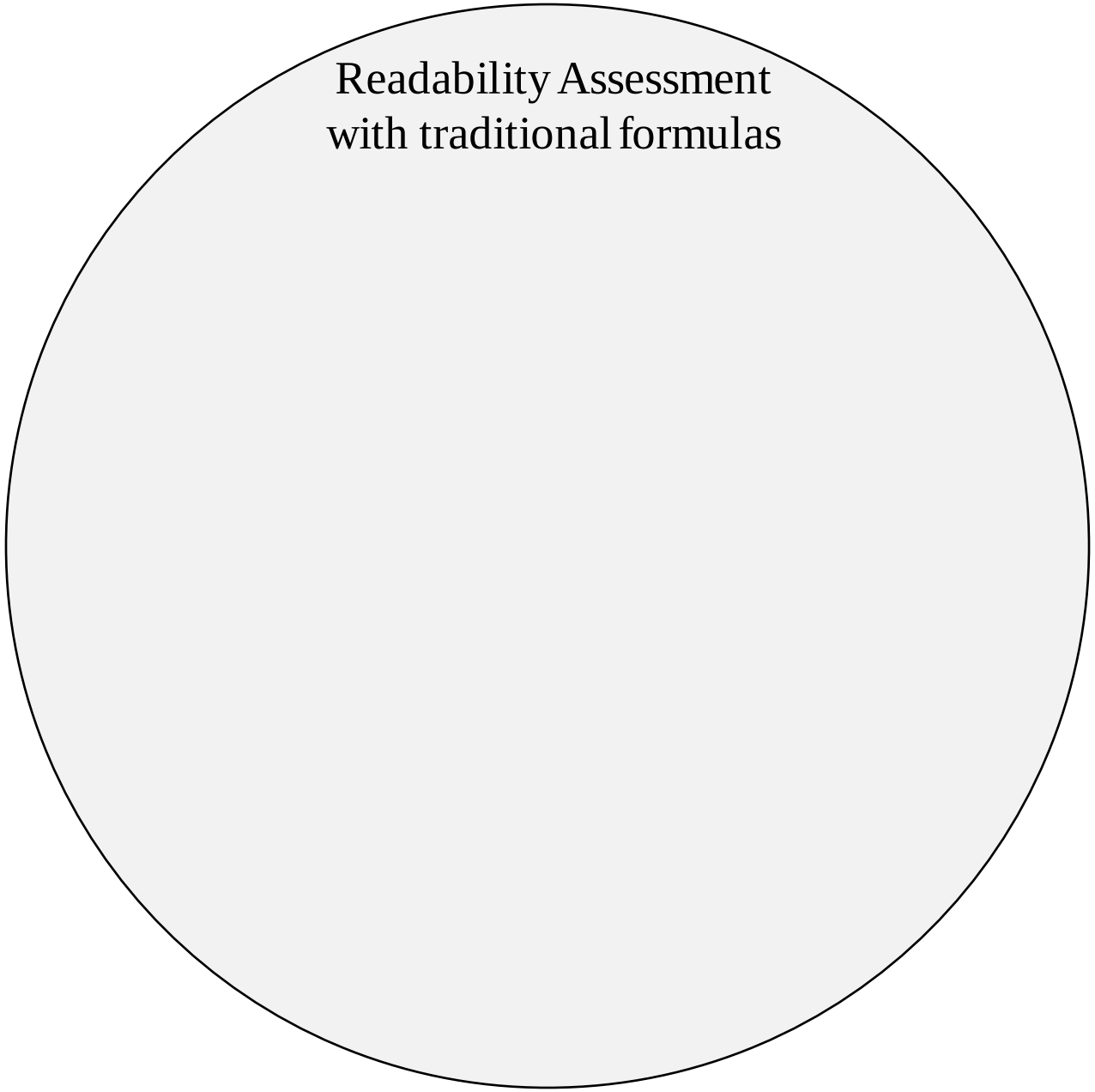
Current trends in readability assessment



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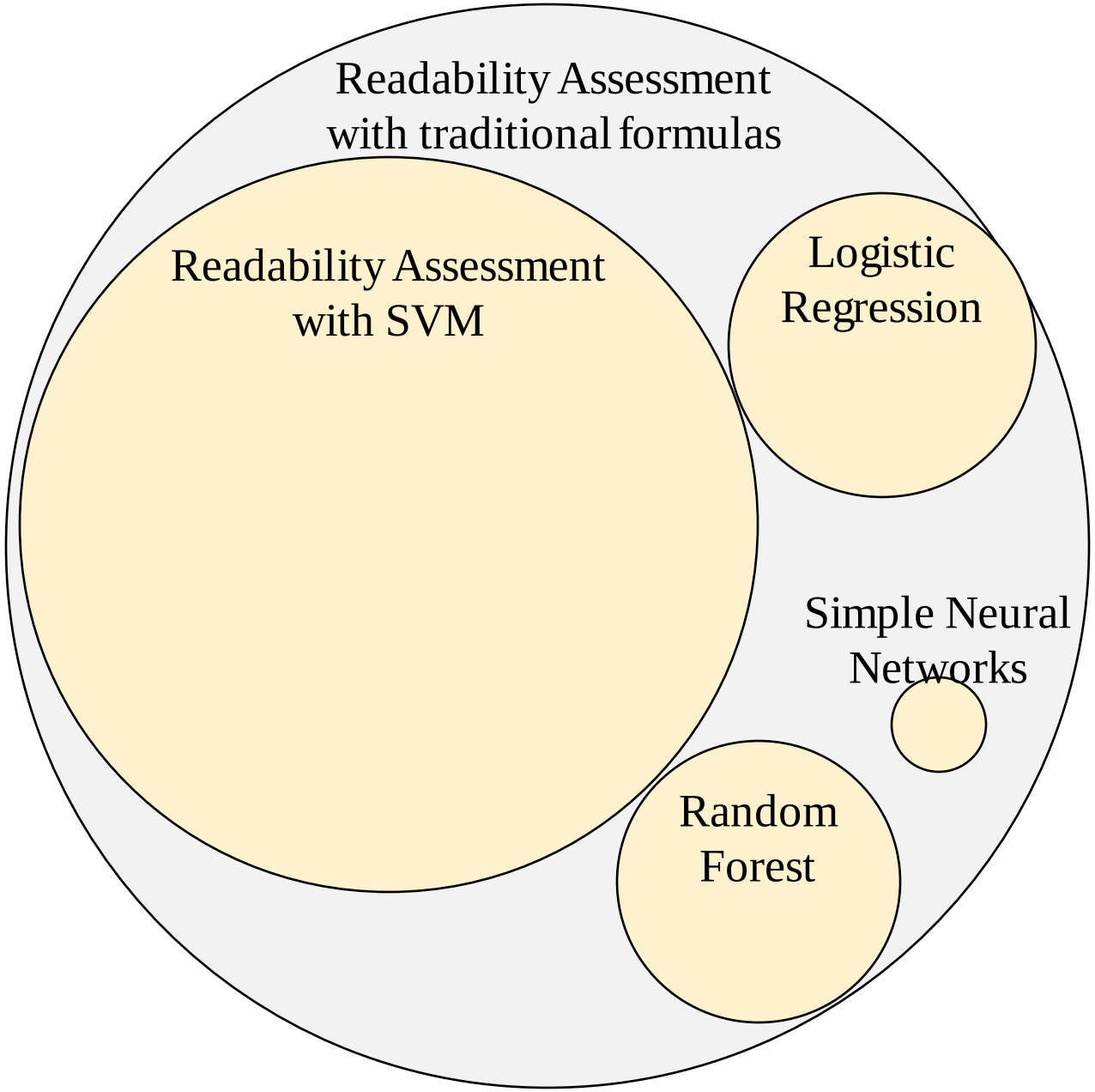

Traditional machine learning methods based on a fixed selection of features (Hansen et al., 2021)

Few consider neural networks in the first place.

Very difficult to find models use complex deep le arning architectures (attention, transformer, etc) not based on simple neural networks.

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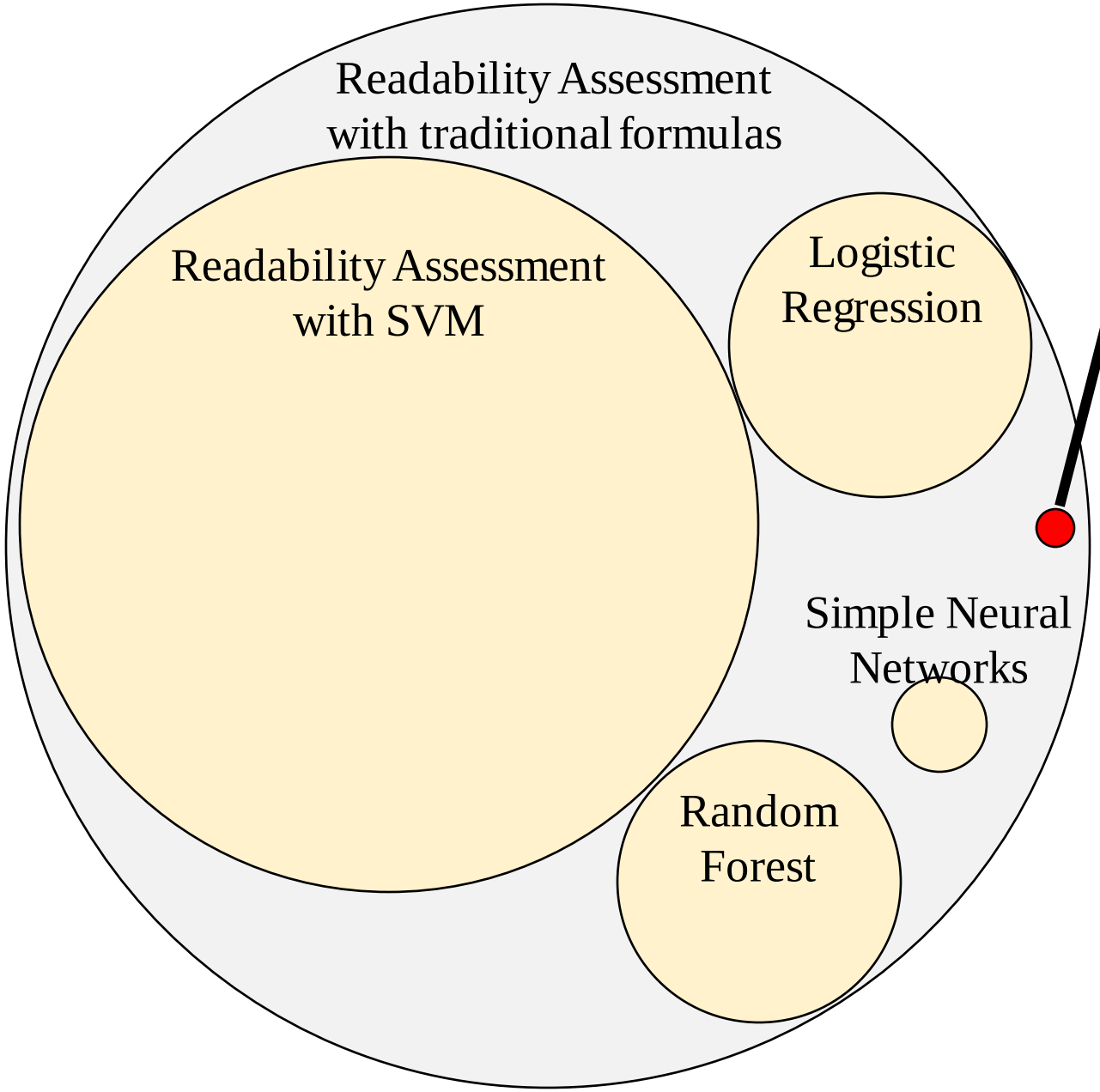

BERT on readability assessment show very good performance! (Martinc et al., 2021)

Raised SOTA accuracy by 4%.

BUT not as strong on a smaller dataset.

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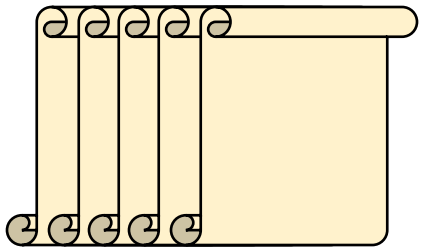
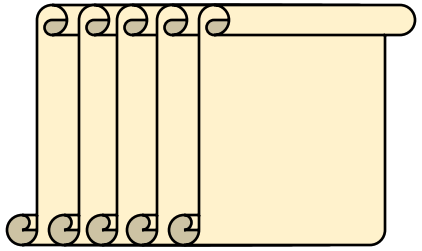
Datasets we used



WeeBit

- Most popular and the largest
- Obtained directly from authors

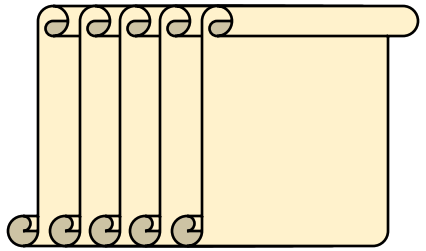
- 5 readability classes
- 625 items/class



OneStopEnglish

- Aligned passages (paraphrased)
- Publicly-available on Github

- 3 readability classes
- 189 items/class



Cambridge

- Cambridge English Exam Levels (CEFR)
- Publicly-available

- 5 readability classes
- 60 items/class

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Datasets we used

	BERT (Martinc et al., 2021)	SVM (Xia et al., 2016)	SVM + SMO (Vajjal and Lucic, 2018)
WeeBit	0.857	0.803	N/A
OneStopEnglish	0.674	N/A	0.781
Cambridge	N/A	0.786	N/A

WeeBit

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Why the low accuracies and inconsistency on smaller datasets?

1. Only BERT was applied to readability assessment.
2. No additional measures done to supply the model with more linguistic information.

- 5 readability classes
- 60 items/class

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Our contributions

1. We numerically represent certain linguistic properties pertaining to advanced semantics.

2. We develop a large-scale, openly available 255 features extraction Python toolkit (which is highly scarce in RA). We name the software LingFeat.

3. We conduct wide searches and parametrizations on transformers and traditional ML for RA use.

4. We develop hybrid models for SOTA and robustness on small datasets.

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3. We conduct wide searches and parametrizations on transformers and traditional ML for RA use.

4. We develop hybrid models for SOTA and robustness on small datasets.

BERT-GB-T1 hybrid beats the SOTA accuracy with only 30% of the original dataset

RoBERTa-RF-T1 achieves 99% accuracy, 20.3% higher than the previous SOTA

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All Features

255 features

Advanced Semantics Syntactic Shallow Traditional

Lexico-Semantic Discourse

Part-of-Speech Features

Variation Ratio Features

Entity Grid Features

Psycholinguistic Features

Entity Density Features

Type Token Ratio Features

Tree Structure Features

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Search on neural model & traditional ML model

Corpus		BERT	RoBERTa	BART	XLNet
WeeBit	Accuracy	0.893	**0.900**	0.889	0.881
	F1	0.893	**0.900**	0.889	0.880
	Precision	0.896	**0.902**	0.892	0.881
	Recall	0.896	**0.902**	0.892	0.881
	QWK	0.966	**0.970**	0.963	0.966
OneStopE	Accuracy	0.801	0.965	**0.968**	0.804
	F1	0.793	0.965	**0.968**	0.794
	Precision	0.815	0.968	**0.970**	0.810
	Recall	0.814	0.968	**0.970**	0.810
	QWK	0.840	0.942	**0.952**	0.845
Cambridge	Accuracy	0.573	**0.680**	0.620	0.573
	F1	0.517	**0.658**	0.598	0.554
	Precision	0.528	**0.693**	0.643	0.591
	Recall	0.525	**0.693**	0.643	0.591
	QWK	0.809	**0.881**	0.835	0.832

- RoBERTa shows overall best performance on readability assessment datasets.
- All transformers showed sudden performance drops on Cambridge dataset.
- Most likely due to the varying pre-training steps of each transformer.

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Search on neural model & traditional ML model

- All four models show worse results than transformers on all datasets/metrics.
- SVM was better for the biggest dataset but not for the smallest dataset.
- Consistent, quite-expectable performance across different dataset sizes.

Corpus		SVM	RandomF	XGBoost	LogR
WeeBit	Accuracy	**0.679**	0.638	0.638	0.622
	F1	**0.672**	0.626	0.627	0.615
	Precision	**0.696**	0.645	0.656	0.676
	Recall	**0.679**	0.638	0.638	0.622
	QWK	**0.716**	0.703	0.692	0.640
OneStopE	Accuracy	0.737	0.709	0.719	**0.778**
	F1	0.730	0.706	0.701	**0.770**
	Precision	0.751	0.726	0.734	**0.778**
	Recall	0.737	0.709	0.719	**0.778**
	QWK	0.400	0.434	0.363	**0.486**
Cambridge	Accuracy	0.627	0.673	**0.685**	0.680
	F1	0.613	0.663	**0.681**	0.657
	Precision	0.660	0.696	**0.701**	0.694
	Recall	0.627	0.673	**0.674**	0.680
	QWK	0.857	0.880	**0.852**	0.856

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Feature sets

Set	Features	LogR	SVM
T1	**AdSem + Disco + Synta + LxSem + ShaTr**	**0.622**	**0.679**
T2	Disco + Synta + LxSem + ShaTr	0.528	0.546
T3	AdSem + Synta + LxSem + ShaTr	0.591	0.582
H1	**AdSem + Disco**	**0.463**	**0.513**
L1	Synta + LxSem	0.499	0.561
L2	**Set L1 - PhrF**	**0.539**	**0.577**
L3	Set L1 - VarF	0.529	0.551
L4	Set L1 - POSF	0.449	0.551
E1	AdSem + PsyF + WorF + TraF	0.489	0.473
E2	**AdSem + PsyF + WorF**	**0.490**	**0.479**
E3	PsyF + WorF	0.464	0.459
P1	EnDF + ShaF + TrSF + POSF + WorF + PsyF + TraF	0.608	0.633
P2	Set P1 + TraF	0.629	0.638
P3	**Set P2 + VarF**	**0.647**	**0.674**

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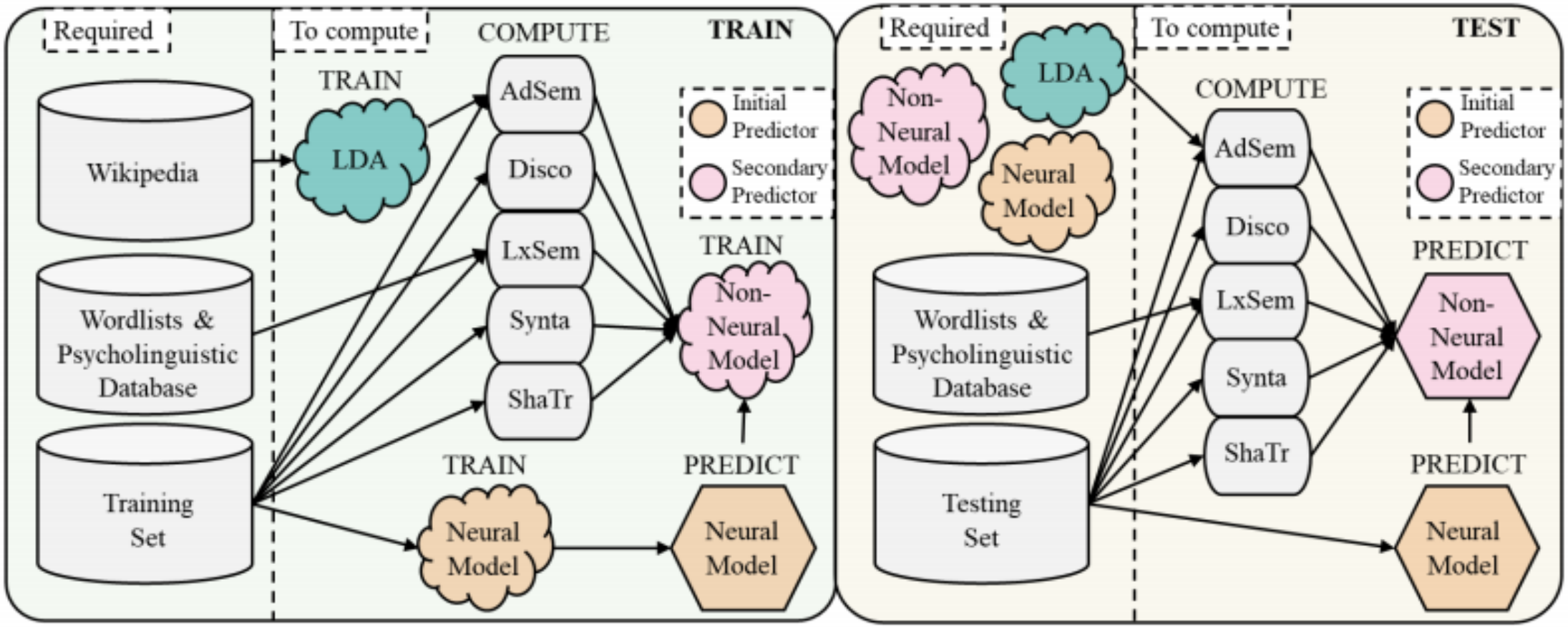
Assembling a hybrid

4 x 4 x 14 = 224 combinations



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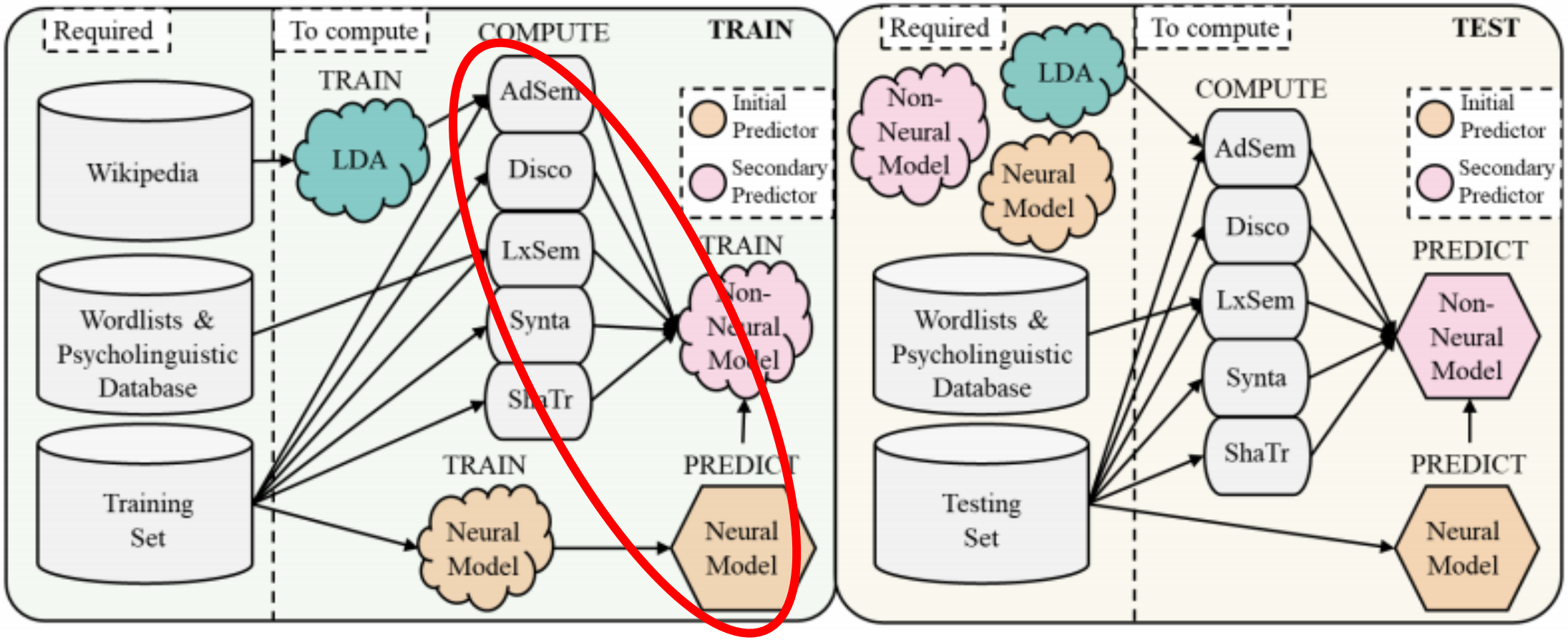

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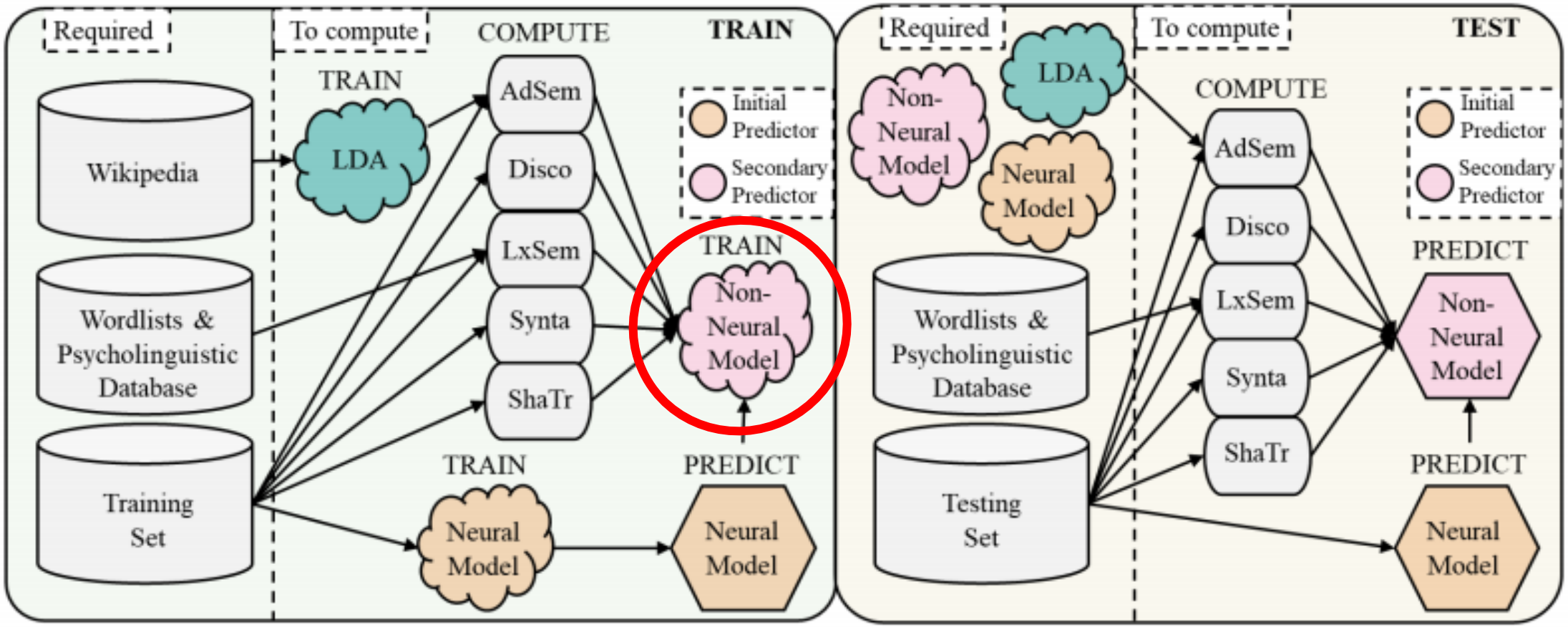

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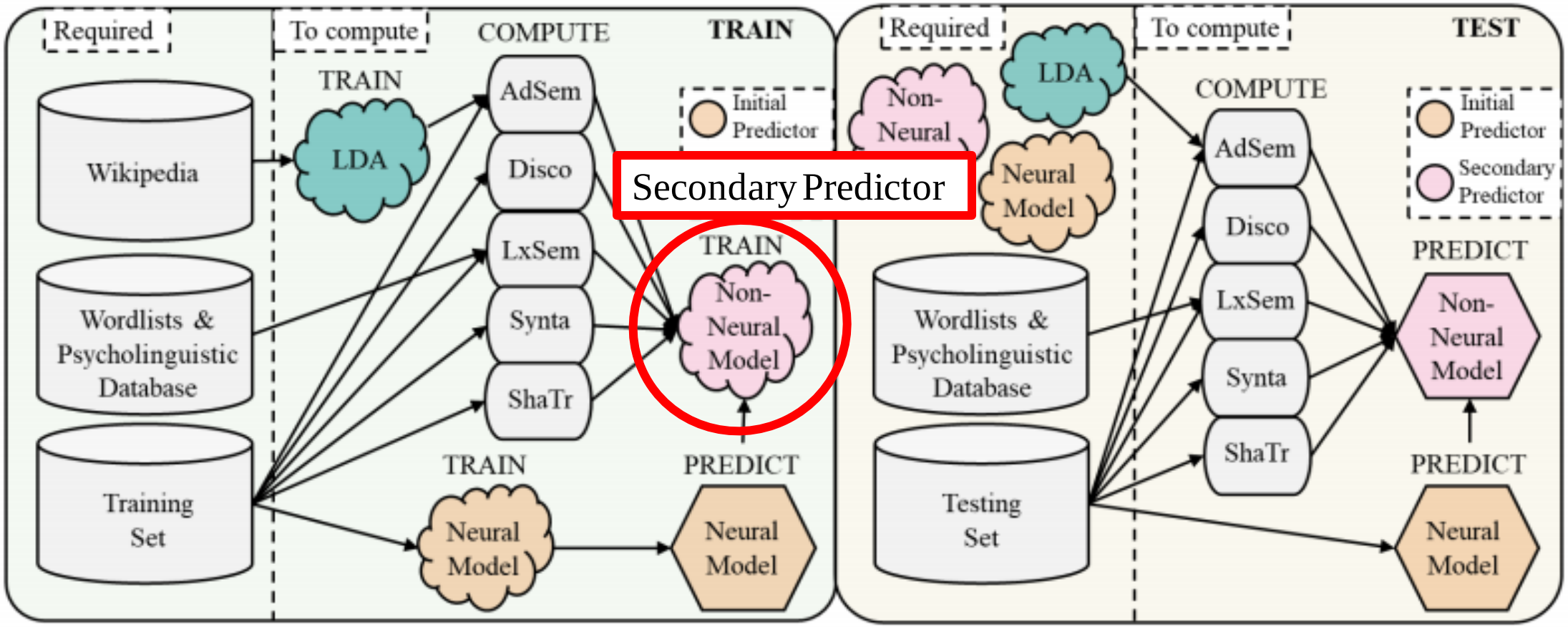

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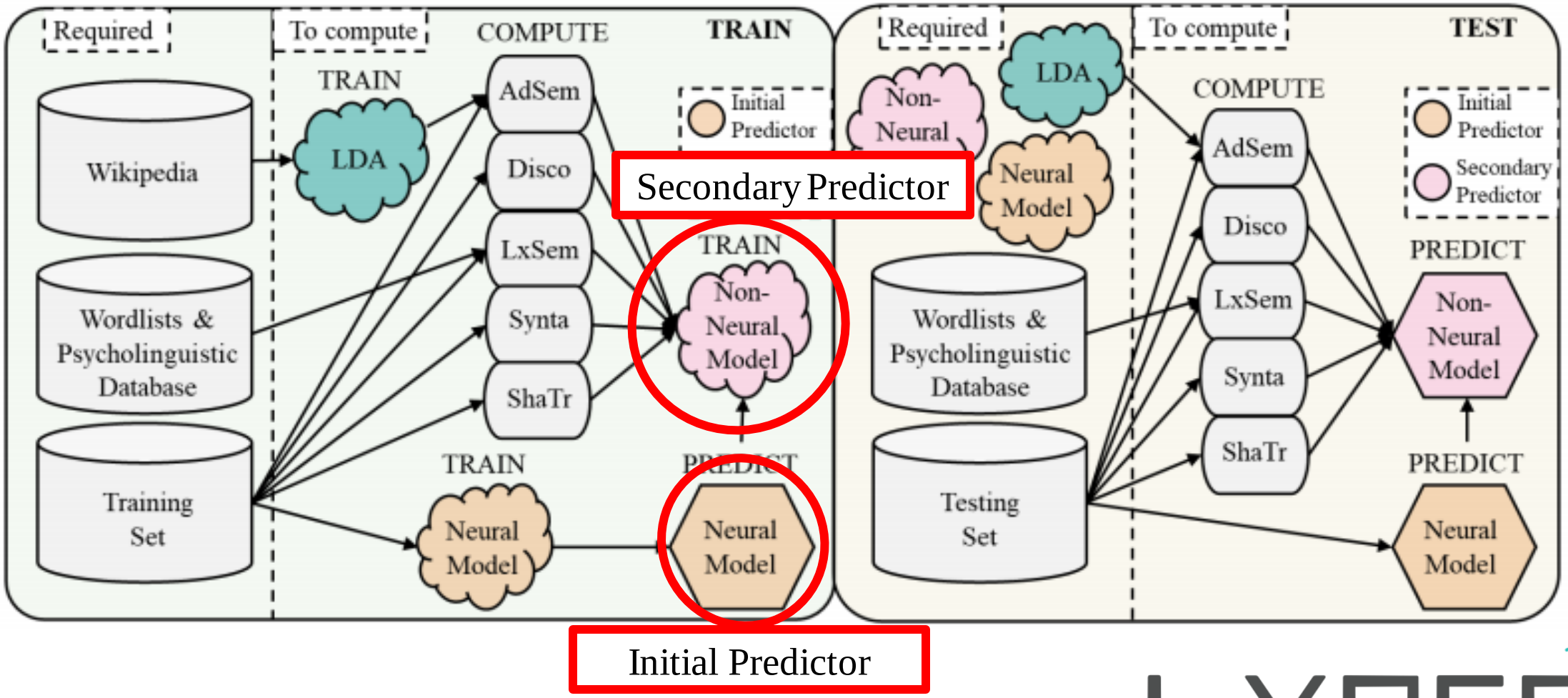

Initial Predictor

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Hybrid model results

Corpus		Model															
		Baselines, Previous Studies				BERT			RoBERTa			BART			XLNet		
		Xia-16	Fili-19	Mar-21		hybrid	Δ	Δ	hybrid	Δ	Δ	hybrid	Δ	Δ	hybrid	Δ	Δ
		SVM	LSTM	BERT	HAN	GB-T1	BERT	GB	RF-T1	RBRT	RF	RF-T1	BART	RF	RF-P3	XLNet	RF
WeeBit	Accuracy	0.803	0.813	**0.857**	0.752	0.895	0.002	0.257	0.902	0.002	0.264	**0.905**	0.016	0.267	0.892	0.011	0.254
	F1	-	-	**0.866**	0.753	0.895	0.002	0.268	0.902	0.002	0.276	**0.905**	0.016	0.279	0.892	0.012	0.266
	Precision	-	-	**0.857**	0.752	0.897	0.001	0.241	0.903	0.001	0.258	**0.905**	0.013	0.260	0.893	0.012	0.248
	Recall	-	-	**0.858**	0.752	0.897	0.001	0.259	0.903	0.001	0.265	**0.904**	0.012	0.266	0.892	0.011	0.254
	QWK	-	-	**0.953**	0.886	0.969	0.001	0.277	0.971	0.001	0.268	**0.968**	0.005	0.265	0.966	0.000	0.263
OneStopE	Accuracy	-	-	0.674	**0.787**	0.982	0.181	0.263	**0.990**	0.025	0.281	0.971	0.003	0.262	0.848	0.044	0.139
	F1	-	-	0.740	**0.798**	0.982	0.189	0.281	**0.995**	0.030	0.289	0.971	0.003	0.265	0.848	0.050	0.142
	Precision	-	-	0.674	**0.787**	0.983	0.168	0.249	**0.995**	0.027	0.269	0.972	0.002	0.246	0.852	0.042	0.126
	Recall	-	-	0.677	**0.789**	0.982	0.168	0.263	**0.996**	0.028	0.287	0.971	0.001	0.262	0.848	0.038	0.139
	QWK	-	-	0.708	**0.825**	0.973	0.133	0.610	**0.996**	0.054	0.562	0.952	0.000	0.518	0.855	0.010	0.369
Cambridge	Accuracy	0.786**	-	-	-	0.687	0.114	0.002	**0.763**	0.083	0.090	0.727	0.107	0.054	0.687	0.114	0.014
	F1	-	-	-	-	0.682	0.165	0.001	**0.752**	0.094	0.089	0.727	0.129	0.064	0.676	0.122	0.013
	Precision	-	-	-	-	0.732	0.204	0.031	**0.792**	0.099	0.096	0.760	0.117	0.064	0.710	0.119	0.014
	Recall	-	-	-	-	0.687	0.162	0.013	**0.753**	0.060	0.080	0.727	0.084	0.054	0.687	0.096	0.014
	QWK	-	-	-	-	0.873	0.064	0.021	**0.919**	0.038	0.039	0.889	0.054	0.009	0.888	0.056	0.008

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WeeBit	Accuracy	0.803	0.813	**0.857**	0.752	0.895	0.002	0.257	0.902	0.002	0.264	**0.905**	0.016	0.267	0.892	0.011	0.254
	F1	-	-	**0.866**	0.753	0.895	0.002	0.268	0.902	0.002	0.276	**0.905**	0.016	0.279	0.892	0.012	0.266
	Precision	-	-	**0.857**	0.752	0.897	0.001	0.241	0.903	0.001	0.258	**0.905**	0.013	0.260	0.893	0.012	0.248
	Recall	-	-	**0.858**	0.752	0.897	0.001	0.259	0.903	0.001	0.265	**0.904**	0.012	0.266	0.892	0.011	0.254
	QWK	-	-	**0.953**	0.886	0.969	0.001	0.277	0.971	0.001	0.268	**0.968**	0.005	0.265	0.966	0.000	0.263
OneStopE	Accuracy	-	-	0.674	**0.787**	0.982	0.181	0.263	**0.990**	0.025	0.281	0.971	0.003	0.262	0.848	0.044	0.139
	F1	-	-	0.740	**0.798**	0.982	0.189	0.281	**0.995**	0.030	0.289	0.971	0.003	0.265	0.848	0.050	0.142
	Precision	-	-	0.674	**0.787**	0.983	0.168	0.249	**0.995**	0.027	0.269	0.972	0.002	0.246	0.852	0.042	0.126
	Recall	-	-	0.677	**0.789**	0.982	0.168	0.263	**0.996**	0.028	0.287	0.971	0.001	0.262	0.848	0.038	0.139
	QWK	-	-	0.708	**0.825**	0.973	0.133	0.610	**0.996**	0.054	0.562	0.952	0.000	0.518	0.855	0.010	0.369
Cambridge	Accuracy	0.786**	-	-	-	0.687	0.114	0.002	**0.763**	0.083	0.090	0.727	0.107	0.054	0.687	0.114	0.014
	F1	-	-	-	-	0.682	0.165	0.001	**0.752**	0.094	0.089	0.727	0.129	0.064	0.676	0.122	0.013
	Precision	-	-	-	-	0.732	0.204	0.031	**0.792**	0.099	0.096	0.760	0.117	0.064	0.710	0.119	0.014
	Recall	-	-	-	-	0.687	0.162	0.013	**0.753**	0.060	0.080	0.727	0.084	0.054	0.687	0.096	0.014
	QWK	-	-	-	-	0.873	0.064	0.021	**0.919**	0.038	0.039	0.889	0.054	0.009	0.888	0.056	0.008

Hybrid model results

Corpus		Model															
		Baselines, Previous Studies				BERT			RoBERTa			BART			XLNet		
		Xia-16	Fili-19	Mar-21		hybrid	Δ	Δ	hybrid	Δ	Δ	hybrid	Δ	Δ	hybrid	Δ	Δ
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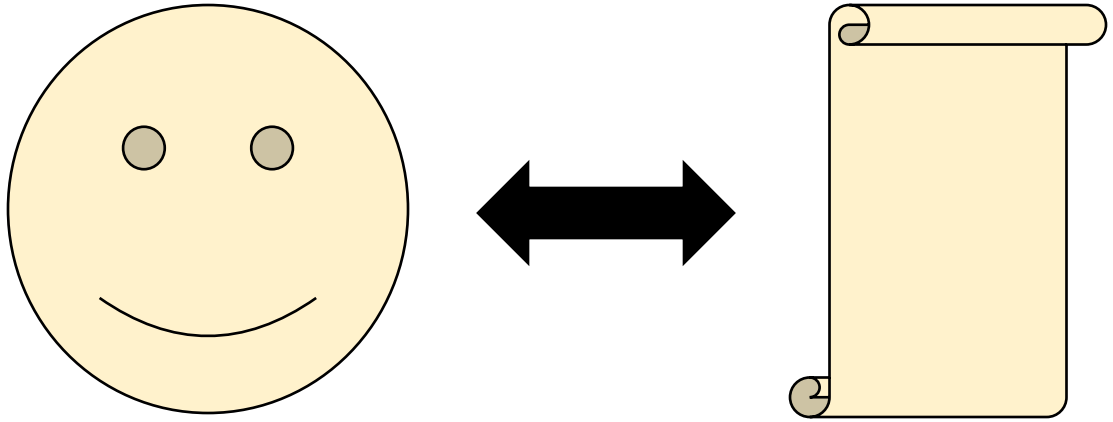
Conclusion

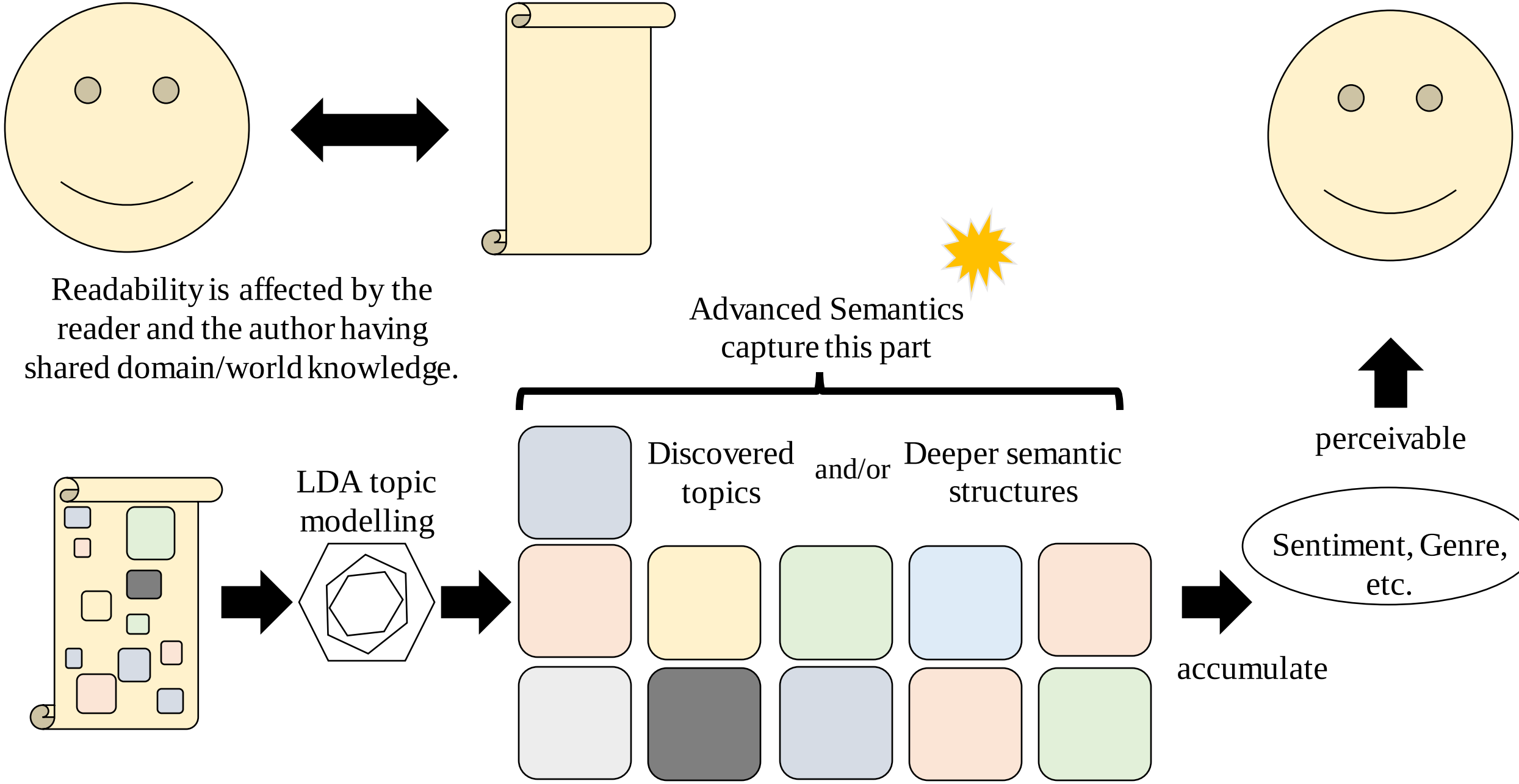
Appendix

Advanced Semantics

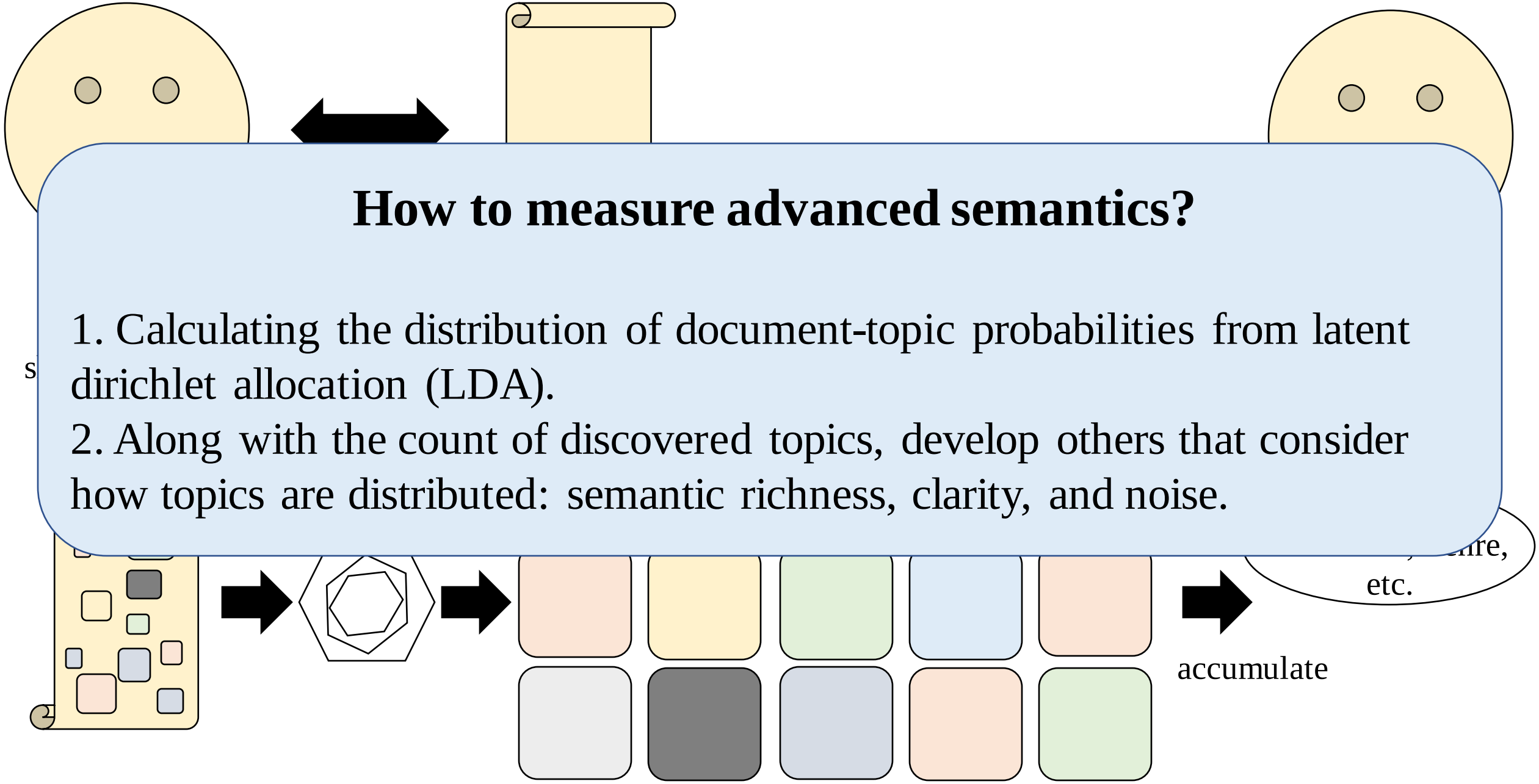


Readability is affected by the reader and the author having shared domain/world knowledge.

Advanced Semantics



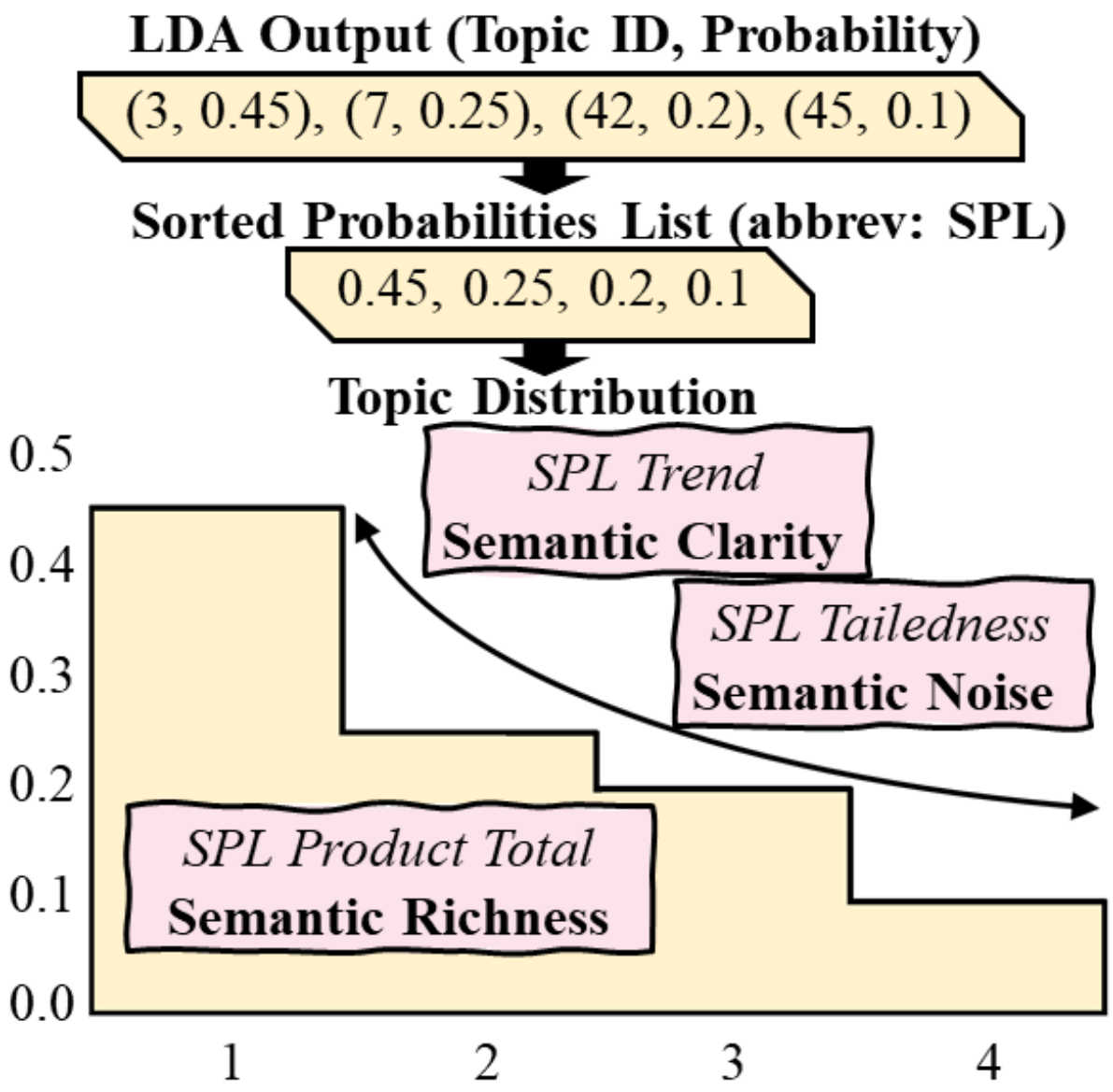
Advanced Semantics



How to measure advanced semantics?

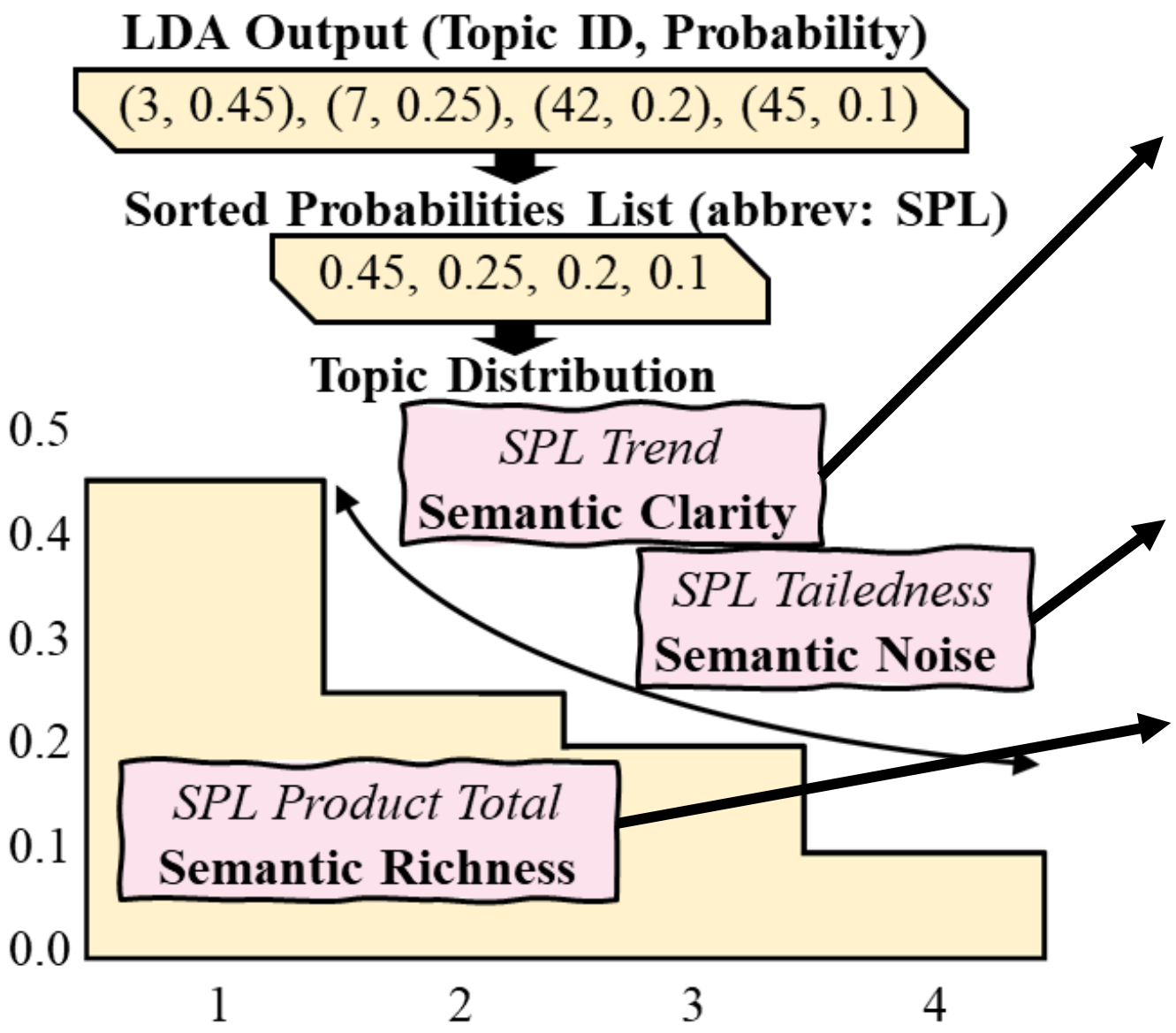
1. Calculating the distribution of document-topic probabilities from latent dirichlet allocation (LDA).
2. Along with the count of discovered topics, develop others that consider how topics are distributed: semantic richness, clarity, and noise.

Advanced Semantics



LXPER

Advanced Semantics



Semantic Clarity: how the probabilities list is skewed towards the largest topic.

$$\text{Semantic Clarity} = \frac{1}{n} \cdot \sum_{i=1}^{n} (max(p) - p_i)$$

Semantic Noise: how less-important topics are, also the "tailedness".

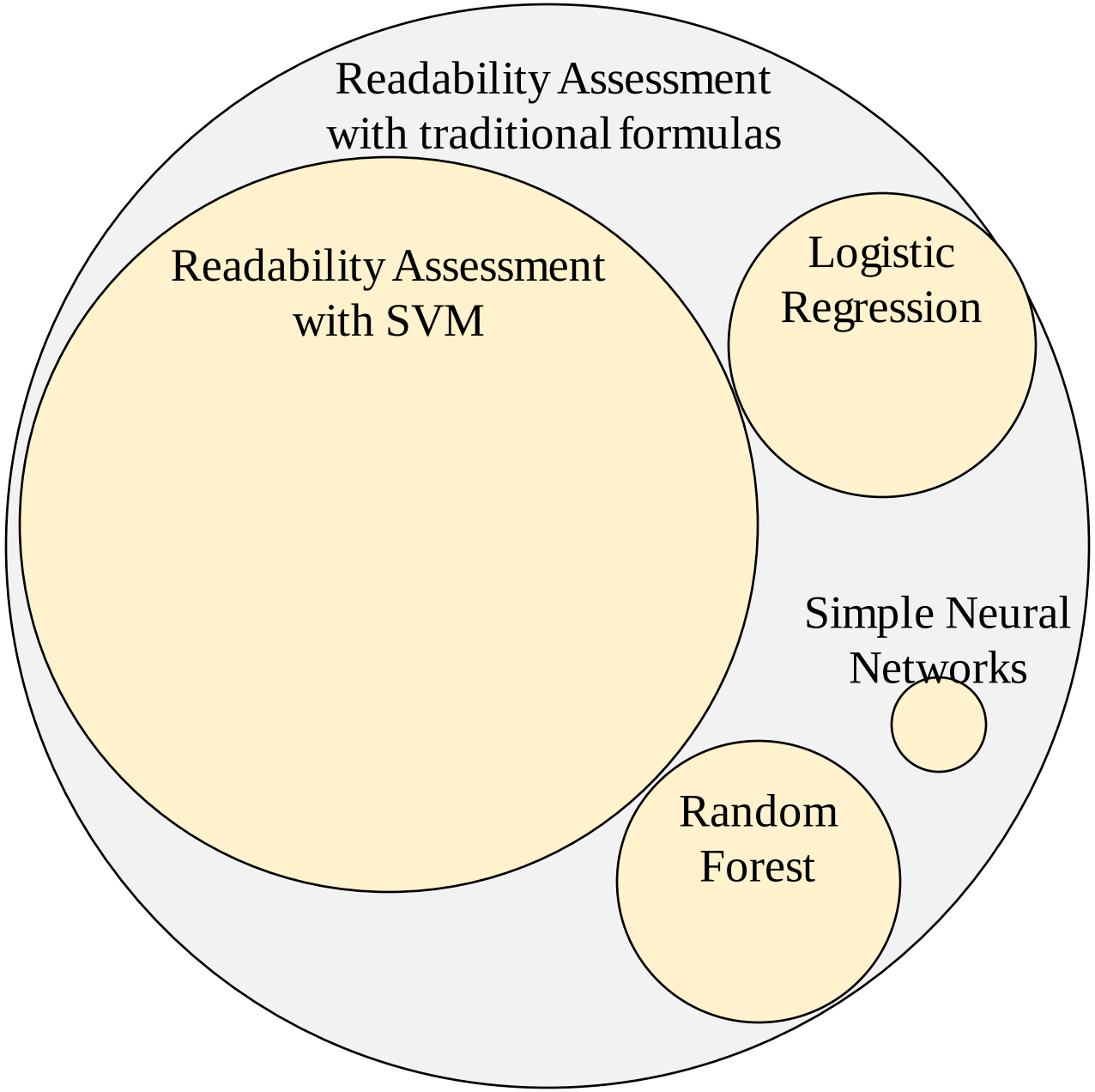
$$\text{Semantic Noise} = n \cdot \frac{\sum_{i=1}^{n} (p_i - \bar{p})^4}{(\sum_{i=1}^{n} (p_i - \bar{p})^2)^2}$$

Semantic Richness: how "largely" populated the topic are.

$$\text{Semantic Richness} = \sum_{i=1}^{n} p_i \cdot i$$

LXPER

Current trends in readability assessment

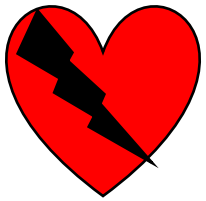


Traditional machine learning methods based on a fixed selection of features (Hansen et al., 2021)

Few consider neural networks in the first place.

Very difficult to find models use complex deep le arning architectures (attention, transformer, etc) not based on simple neural networks.

No Match!

Transformers  Readability Assessment

LXPER

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